

Calibration as Linkage Integrity

Does a knowledge base's rigor survive contact with a reasoner?

Submission to the FLF Epistemic Case Study Competition ("Epistack")

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Materials: a written argument (this document), a runnable instrument (`epistack_eval.py`), and its run outputs across three case shapes and a real large base, on two models, all in the linked repository.

I sent my original submission for the early feedback round and received some important feedback from Oly. Since then I have taken that direction and kept building on it. I have substantially expanded my work. It is now an operational instrument that runs on a real 26-claim knowledge base, and it has a third probe, a retracted-claim provenance test. The compute support from FLF covered my additional runs. Though the core idea is unchanged, what has changed is that it is now tested on a serious artifact, with findings reported at their true size.

What I am submitting currently is a working detector which measures whether a knowledge base's rigor survives a reasoner, and it has caught a real failure on a real base. But I want to specifically call out that the detector is just the first step. I aim to make the instrument a complete Epistack Eval Standard, a single score that any base can be measured against, and then a repair tool that fixes the failures it finds. I built and proved the detector here, while the Standard and the repair tool are where it goes and what I want to build next. I lay them out at the end.

The core submission

The real test of the rigor of a knowledge base is how well it survives contact with a reasoner. Ideally the reasoner leaning only on the base should be as confident as the evidence in the base warrants. Any more or less confidence is a failure of the base and not a failure of the reader. My submission is a way to measure that deviation, and an instrument that does it on real cases.

One thing that I want to call out upfront: The knowledge base is an input to this instrument and is not a part of it. So it can be pointed at anyone's knowledge base and not just mine. That means any other entrants, or any base FLF itself wants to check. That makes it a shared way to test whether a base's rigor actually holds when a reasoner is leaning on it, and it is why I think the value this brings is the instrument anyone can use and not just my results.

I am showing this on all three cases of different shape. One is contested, one is settled, and the third is ill-posed. I used the same instrument, with two different models (Sonnet and Opus), to ensure it works as infrastructure. A method that only works on one case is not yet infrastructure. I then went further and ran the same instrument on a real, large 26-claim knowledge base built from deep research, to see whether it holds on a serious artifact and not just starter examples. It does, and that is the result I am adding here.

The problem

The true test of the rigor of a knowledge base lies in the confidence of the reasoner. A base can have a very well organised structure and confident-sounding claims and still leave the reasoner who leans on it believing things the evidence does not support. That gap is invisible when we only inspect the artifact, because the failure shows up somewhere else, in the reasoner who used it.

Why this is the layer that matters

Input-side inspection can miss the effect on a reasoner. Measuring the linkage integrity is the key here. Structure and provenance still matter on the input side, and they certify the quality of what goes in. But even with them, the result may still lead to the reasoner double-counting correlated evidence, missing the one choice the answer actually hinges on, or leaning on a claim that has been withdrawn. Provenance is a good example of why the output side is needed: it matters as an input-side check, but whether a reasoner actually respects it is something you only see by testing what the base does to a reasoner, which is exactly what my retracted-claim probe does. That is why the base has to be tested by what it does to a reasoner, not only by how it looks.

I have put in effort to check and build an infrastructure to ensure that linkages actually hold, from the output side.

Is it beating off-the-shelf deep research? In fact it is not competing with deep research; in terms of layers, it sits one layer above it. Whether the artifact the reasoner tries to lean on is actually rigorous or not is not reflected

from the deep research that handed the reasoner the artifact, the set of claims, and the knowledge base. My instrument does exactly that. You can even run deep research to build the base, then point the instrument at it to see where a reasoner ends up more or less confident than the evidence warrants, which claims are load-bearing, and where correlated evidence gets double-counted. So deep research can produce the artifact; my instrument measures whether the artifact survives contact with the reasoner. To finish the loop, that is a very important gap that deep research doesn't fill but my instrument does.

The instrument

`epistack_eval.py` is a dependency-light script that runs the evaluation. It sends real reasoners, LLM personas, through the knowledge base, and it runs in one command from the linked repository. It is not a mockup with hand-written answers; the reasoning comes from a real model, so the result is not baked in.

The evaluation is one script with three probes:

1. a linkage-integrity test,
2. a behavioral double-counting test, and
3. a retracted-claim provenance test (on the large base), which checks whether a reasoner leans on evidence that has been withdrawn.

Each is designed to measure a different failure.

The linkage test measures all three constraints it applies:

1. reasoning only from the base,
2. updating on a held-out claim it never saw, and
3. holding up under a counter-argument.

The first is enforced by instruction. The next two are measured directly. It then also reports the objective metric, calibration gained per claim read.

The instrument runs each test through three reasoner personas that differ only in the starting belief. One leans toward a lab origin, one is neutral, and one leans toward zoonosis. The three priors spanning the range is the smallest set that tests what matters here, whether the base claims pull the reasoner toward the warranted answer irrespective of where they start. A richer set of personas that vary reasoning style, not just the starting prior, for example one that over-weights vivid specific evidence or one that treats every listed claim as independent, would test more failure modes. That would be my next obvious step.

How it scores

For each persona it records the confidence they end up with, how wide their uncertainty is, whether they were able to identify the load-bearing crux, whether they double-counted correlated clues, and whether their confidence survived the counter-argument. It then compares that against the warranted shape from the public record: the structure of the disagreement, which claims are load-bearing, which clues are correlated, and how wide uncertainty has to be.

For a contested question there is no single true answer to calibrate against, so the instrument does not use one. The target is not a probability. It is the structure of the disagreement, which is a fact about the debate, not a verdict on it.

The three cases

The same instrument runs on three differently-shaped starter cases, and then on a real, large base. Only the knowledge base and the warranted shape change.

- COVID is contested. Wide uncertainty is warranted.
- Black holes is settled. Confident-safe is warranted, and the failure to catch is false balance.
- Eggs is ill-posed. The failure is a confident universal answer instead of catching that the effect depends on who you are.

The first case, COVID, is a genuinely contested question with an unusually complete public record. Six analysts spanned twenty-three orders of magnitude, neutral analysts differed fiftyfold, superforecasters landed near 75/25 for a natural origin, and the public polls around two-thirds for a lab origin. The facts are largely agreed; the disagreement is driven by the reference-class choice, not the facts. That gap between confident opinion and the careful-but-uncertain read is the kind of thing the instrument is built to detect.

What the evaluation found: the headline result on a real large base

I built a real, large knowledge base the way a serious investigator and reasoner would, and then tested it. Here is what I did, in order:

- I ran deep research on the question of COVID origins on two independent models.
- I reconciled the two outputs into a single 26-claim evidence ledger.
- I then stripped the ledger into a flat, unflagged claim list, holding back the correlation structure and strengths separately as a hidden answer key.
- I ran the instrument on the unflagged list.

This directly answered whether this beats off-the-shelf deep research. Deep research built the base, and the instrument tested whether that base's rigor survives a reasoner.

The instrument caught a double-counting failure in the base, reproducibly. The base contains a cluster of five claims that all reduce to one underlying fact, that the relevant lab is in Wuhan, the city where the COVID outbreak began. When those five correlated claims are shown to a reasoner as separate strikes, and not flagged as one shared cause, the reasoner moved further toward a lab origin. On the stronger model this effect averaged about 0.12 across seven separate runs, ranging 0.09 to 0.14. On the lighter model (Sonnet) it averaged about 0.08, ranging 0.06 to 0.11. It was positive in every run on both models, and on every persona. The stronger model showed the larger effect on average, though the two ranges overlap, and in one of seven paired runs the lighter model was slightly higher. The measurement is coarse: models answer in roughly 0.05 steps, so with three personas one persona moving a single step shifts the run mean by about 0.017. The direction is robust; the magnitude is noisy, and I report it that way. The correlation had been removed beforehand, and the instrument, reading the unflagged version, independently caught that a reasoner still double-counts it.

To be clear about what was tested: this probe uses the base's real five-claim Wuhan cluster, not the smaller three-clue probe used on the starter COVID case (shown later, under the starter shapes), which is why the effect comes out clearer here. A reviewer can see exactly which claims are involved, and all fourteen run outputs (seven per model) are in the repository.

The other measures held consistently across runs:

- Reasoners transferred reasoning to the held-out claim every time on both models.
- Crux localization stayed model-dependent, more reliable on the stronger (Opus) model in most runs while patchy on the lighter (Sonnet).
- Warranted-stance survival was also model-dependent. Opus held its ground against a confident lab-leak counter-argument, and Sonnet sometimes caved.
- The calibration-per-claim metric stayed small and close to zero on this base, and reasoners tended to land a little below the warranted zoonosis band.

I am reporting the small numbers as small and the stable numbers as stable.

The point here is not the exact size of the effect, but that a real, large, expert-grade knowledge base, one that is already carefully de-correlated, still led a reasoner reading its unflagged claims to double-count. Flagging correlations in a base does not cover the reasoner who reads the claims. That is a failure that does not come out purely by inspecting the base, and it is exactly what an output-side instrument is for.

The retracted-claim probe: does a reasoner lean on withdrawn evidence

It is possible that a base carries a claim that was later retracted. A reasoner reading the base needs to notice the retraction and shouldn't lean on the withdrawn evidence. So I built a third probe on the large base to test that.

I planted one genuinely retracted claim as if it were a live evidence for a lab origin. The 2020 preprint that reported HIV-1 like insertions in the spike protein and was later withdrawn by its own authors after the claimed similarities were shown to be too generic. I showed it to the reasoner two ways, once plainly and once carrying an explicit note that it had been retracted and is not a valid evidence. What I want to measure here is how much belief still moves when the claim is labelled as retracted.

Across the three runs per model, both models correctly discounted the retracted claim. Shown live, they moved toward a lab origin by about 0.04 to 0.10. Shown with the retraction, that movement essentially vanished. The mean residual move was 0.00 on the stronger Opus model and about minus 0.01 on the lighter Sonnet one.

There are two limitations here. The retraction here was explicitly labelled, so this shows models honor a clear retraction, not that they would catch a quietly withdrawn claim with no marker. The lighter Sonnet model

returned unparseable output on a few cells, so its numbers rest on fewer observations than the stronger model's. This is a provenance check the instrument can now run on any base, and the next step is a subtler version where the retraction has to be reasoned out, not read off a label. That version has to use a synthetic claim the model has never seen, with the retraction cue living inside the base itself, so the reasoner cannot pass it from training memory. Otherwise the knowledge mask leaks and the test measures recall, not provenance reasoning.

It also travels across three starter shapes

The headline above is the large base. The same instrument also runs on three smaller, differently-shaped starter cases, which is how I checked it travels rather than working on one artifact. The effects here are smaller, and I report them at that size.

Two results held across every run, and were run on both the Sonnet and Opus models. Reasoner personas transferred reasoning to the held-out claim, updating in the correct direction, by a sensible amount, and for a reason that used the claim. Their confidence held under a direct counter-argument, and it did not collapse. Both were measured by the instrument.

The crux localization is model-dependent. For COVID it was patchy on the lighter Sonnet model (0.67, and that too kept moving between different runs) and very reliable on the stronger Opus model (1.0). Held-out transfer and confidence survival were 1.0 across models. This led me to understand that crux localization is fragile and model-dependent, which is exactly what the instrument aims to show.

The double-counting effect was small: 0.05 on the lighter Sonnet, and 0.02, not firing, on Opus. Reasoners moved a little further toward a lab origin when correlated clues were shown separately as compared to when flagged as one, which was both modest and fading on Opus.

Across the settled and ill-posed cases

The settled case, black holes, gave the clearest positive result. On the stronger Opus model, calibration gained per claim read was about 0.11, and mean calibration increase about 0.57, with the most skeptical persona gaining most (0.75). Each persona moved to confident-safe, and none fell for false balance when the counter-argument pushed them to doubt a settled answer.

There is another important finding, that the metric is model-dependent. On the contested COVID case it was slightly negative on the lighter Sonnet model (minus 0.01) and slightly positive on the stronger Opus one (0.005). On settled black holes it was clearly positive on both, strongest on the stronger Opus model (0.11). It pointed to the fact that a better reasoner extracts more warranted calibration from the same base.

The eggs case is ill-posed, and it behaved as predicted. On the lighter Sonnet model, crux localization was 0.0: the reasoners missed that the effect is heterogeneous across people, and answered as posed. On the stronger Opus model it was 1.0, meaning they caught it. The metric stayed small and honest either way (0.004 to 0.009).

One caveat: the held-out transfer for black holes scored 0.0 on both the models. The held-out claim reinforced an answer that is already at zero, so there was actually no room left to deviate, and the instrument reported no update as there was no transfer. I note it rather than dress it up.

How this compounds

The knowledge base was built by a pipeline that compounds. Two independent deep-research passes, reconciled into one ledger that caught what each pass missed, and then stripped into a testable base and a separate answer key. Each stage added value. More passes, more models, or more scrutiny would strengthen the base further without changing the method. The base gets more trustworthy as more work is applied to it.

The output of the instrument is structured. It is not a narrative. It reports, per claim and per reasoner, crux localization, double-counting, the calibration metric, and stance survival, as machine-readable results. This can be picked up by anyone and extended for their use.

As I mentioned earlier, the knowledge base is an input, so the instrument can point at any base. FLF can run it across other entrants submissions to see whose linkages actually hold good. The large base run is proof, the base it tested was generated externally by deep research and not something I wrote.

The whole system keeps compounding and checking itself. Deep research builds a base, reconciliation improves it, and, most importantly, the instrument audits whether that improvement actually protects a reasoner. When it did not protect, the instrument said so clearly.

What it does not show is that a single human reasoner's reasoning improves turn over turn. That is not what I tested or claimed. What compounds here is the base and the tooling, across models, passes, and people.

What it does not show

These are its limitations:

1. These are LLM personas across a few runs, and not a cohort of humans. This means the results are more pointing than proving, with the named models.
2. The key numbers kept moving on different runs and between models, so I report stability, instability, and model-dependence instead of single figures.
3. The black-holes and eggs knowledge bases are compact, defensible starters, meant to show the instrument travels, not authoritative verdicts on those cases.
4. Effort is proxied by claims read, and not real human time. More runs, larger bases, and more constraints are the natural next steps, and the comparative direction they enable is laid out in Where this goes.
5. Provenance now gets directly tested. The retracted claim probe checks whether a reasoner leans on withdrawn evidence or not. So far it shows models honoring an explicit retraction, a harder unlabelled version is the next step.
6. The two models are not matched on sampling temperature in the double-counting probe. Sonnet runs at 0.7; Opus does not accept a temperature parameter, so it runs at its default. This is constant across every run, and I name it rather than hide it.

Other constraints I did not build

This instrument tested 3 linkage constraints, a double-counting probe and lastly a retracted claim probe (Part C). A few others did cross my mind that would have strengthened it, which I didn't build to keep the instrument sharp and shippable considering the time constraint:

1. Order sensitivity: present the same evidence in a different order and see if the conclusion changes. I left it out to keep runs comparable.
2. A distractor: add an irrelevant but plausible claim and see if it moves the needle.
3. Consistency across restarts: run the same persona twice and measure variance, which my results suggest would itself be revealing, since the numbers already move run to run.
4. Authority pressure: tell the reasoner most experts think X and see if it caves.
5. Missing-evidence awareness: ask what it would need to resolve the question, where a good answer names the reference-class gap.

The planted retracted claim test was on this list as the most logical and valuable next addition. I built it and it is the Part C above. My next step would be an unlabeled subtler version of it, built on a synthetic claim so the reasoner cannot lean on training memory, which would otherwise break the knowledge mask.

With more time in hand, I would pressure test some of these and add them. I have called and listed them here purposely because knowing which tests would extend and expand the instrument or contain it is my intentional effort to show methodological transparency for this submission.

It matters

This is a working instrument that measures whether a knowledge base's rigor survives a reasoner, on real cases, with reproducible results a reviewer can run. We now have an instrument that can test calibration. That is the difference between an argument about calibration and an instrument that tests it.

Why did I make it

Influencing and leading people from data, to information, to sound decisions under ambiguity and uncertainty at corporate scale is my job in learning design and employee transformation. That is where the instinct comes from, that rigor is only impactful if it survives contact with the recipient. I built this instrument to make that instinct measurable.

Running it

It runs on the following: one file, a command, and an API key. It sends the LLM personas through each case's knowledge base under a knowledge mask. It then applies the three linkage constraints, the double-counting probe, and the retracted-claim probe, and prints the linkage report, the double-counting result, the retracted-claim result, and the calibration-increase-per-reading-effort metric. The knowledge base and the warranted shape are its inputs, so it runs on other cases unchanged. The run outputs are in the repository, and any reviewer can rerun and compare.

Where this goes: from a detector to the Epistack Eval Standard

What i have built is a detector pointed at one base which measures whether a reasoner leaning on that base double counts correlated evidence, misses the crux or leans on a withdrawn evidence. This is a demonstrated result on a real base and it is bounded.

This opens a comparative direction which follows directly from a property the instrument already has. Since base is an input, the instrument runs unchanged across many bases which are built by different people or systems, on the same question. You move from "does this base hold up under a reasoner" to "of every base anyone has built on this question, which hold up and which collapses". That moves this from an instrument to a standard, an Epistack Eval Standard, a single reusable score any base can be measured against (a linkage integrity score), so that bases built by wholly different methods and people become comparable on the important thing, i.e. whether their rigor survives a reasoner. This is the direction Oly pointed at in his early feedback, being interested to see how this gets operationalised and might end up as an Epistack Eval, and it is exactly what I want to build.

The sharpest version of that score is adversarial in nature, which doesnt measure calibration just once. It measures how well a reasoner leaning on a base holds its calibration under sustained, targeted cross examination, and ranks bases by which one produces a reasoner that decays under pressure. A base whose linkages are sound should hold, while a base that only looked sound should come apart when pressed. That metric turns "resolving a contested question" into something measurable across competing structures.

There is a problem that sits between here and there though. To rank bases against each other, the measurement has to be steadier than the differences between the bases, which right now is not the case. My own runs moved from one run to the next, the same base reads from 0.09 to 0.14. That deviation is fine when I report a single range, which is what I do now. It is not fine for ranking, because the gap between 2 bases could be smaller than the wobble inside one base itself. So the core research problem of the comparative version is driving the measurement deviation or noise below the signal between bases. That is some real work, and more runs narrow it slowly. A less coarse metric that aggregates over many probe points instead of differencing 2 coarse outputs could narrow it faster. But it needs to be solved before any ranking means anything.

The second arm that I want to build alongside it is repairing, not just detecting and ranking. The instrument already locates the correlation structure a reasoner is blind to. The repair version uses that to reshape the base, collapse the correlated cluster into one flagged node, and then re-measures to confirm the double-counting actually dropped. That closes the loop from diagnosis to fix to check. The reason it belongs here is that "confirm it dropped" needs a measurement stable enough to trust.

Now, none of this here is built as of now. What is built is a detector and the evidence that the failure it catches is a real one on a real base. This is the direction that evidence opens, the score other work can be measured against, the adversarial version of that score, and the honest obstacle in the way. I am currently submitting the proof point and showing where it leads to.

Appendix: Run Results

Epistack Evaluation: Run Results

Following are the results from running the instrument on the three cases using two models (Sonnet and Opus). It was done using LLM personas, so the results are pointing and not proving.

The main results

Case	Shape	Model	Found the load-bearing crux	Transferred reasoning to held-out claim	Confidence survived counter-argument	Metric (calibration per claim read)
COVID	contested	Sonnet	0.67	1.0	1.0	about -0.01
COVID	contested	Opus	1.0	1.0	1.0	about +0.005
Black holes	settled	Sonnet	1.0	0.0 (see note)	1.0	about +0.09
Black holes	settled	Opus	1.0	0.0 (see note)	1.0	about +0.11
Eggs	ill-posed	Sonnet	0.0	1.0	1.0	about +0.009
Eggs	ill-posed	Opus	1.0	1.0	1.0	about +0.004

What the numbers say

For the settled case of black holes, it gave the clearest positive result (rightfully so). On Opus the mean calibration increase was 0.57, and the most skeptical persona gained the most (0.75). None of the personas fell for false balance when the counter-argument pushed them to doubt a settled answer.

The metric pointed to being model-dependent. It came out cleaner and more positive on the stronger model, and weaker and slightly negative on the lighter one. Which again proved the point that a better reasoner pulls more warranted calibration out of the same base.

Even the crux localization is model-dependent. It was more reliable on Opus and patchy on Sonnet, and it kept moving between different runs.

For the eggs case, it showed its characteristic failure. On the Sonnet model the reasoners missed that the effect is heterogeneous across people, while on Opus they caught it.

The black-holes held-out note

The held-out claim transferred zero on black holes for both models, as it is a settled case. It is a scoring edge. The held-out claim reinforced an answer that was already at zero, so there was no room left to move, and hence the instrument acted correctly in recording no update and no transfer.

Limitations

- 1. It used LLM personas and not a human cohort, though it had multiple runs.
- 2. It scored against the structure of each case.
- 3. The effort metric is proxied by the claims it read.
- 4. The black-holes and eggs knowledge bases are compact, defensible starters, which are meant to show the instrument travels, and not necessarily authoritative verdicts.

Large base (covid_large): the reconciled 26-claim ledger, unflagged

Same instrument, run on a real 26-claim COVID-origins base built from deep research (two models, reconciled), with the correlations and strengths held back as a hidden answer key. Seven runs per model (the full trend table is below).

Measure	Sonnet	Opus
Crux localization	1.0	0.67
Held-out transfer	1.0	1.0
Warranted-stance survival	0.67	1.0
Calibration per claim read	about 0.003	about 0.003
Double-counting effect (real 5-claim Wuhan cluster), 7 runs	mean 0.08, range 0.06 to 0.11	mean 0.12, range 0.09 to 0.14

The double-counting effect is the headline, and it was measured across seven runs per model. On the stronger model it averaged 0.12 (range 0.09 to 0.14); on the lighter model 0.08 (range 0.06 to 0.11). It fired in every one of the fourteen runs and was positive on every persona. The reconciliation had removed this correlation by hand; the instrument, reading the unflagged version, independently caught that a reasoner still double-counts the five Wuhan-cluster claims. Note: Part B here uses this base's real five-claim Wuhan cluster (c3, c20, c21, c22, c23), not a generic probe, so a reviewer can see exactly what was tested.

Trend across seven runs per model (every run counted, none discarded)

mean_double_counting_effect on covid_large, in run order:

Run	Opus	Sonnet	Opus minus Sonnet
1	0.127	0.077	+0.050
2	0.117	0.060	+0.057
3	0.127	0.083	+0.044
4	0.107	0.107	0.000
5	0.107	0.067	+0.040
6	0.137	0.083	+0.054
7	0.093	0.100	-0.007
mean	0.116	0.083	+0.034

mean	0.110	0.082	+0.034
range	0.093 to 0.137	0.060 to 0.107	
sd	0.015	0.017	

What this shows honestly. The effect fired in 14 of 14 runs and was positive on every persona, so the direction is robust. The magnitude is noisy: only 2 of 7 Opus runs land in the 0.12 to 0.13 band an earlier draft of this document claimed, which is why that claim has been replaced with the mean and the full range. The stronger model shows the larger effect on average (+0.034), but the two ranges overlap between 0.093 and 0.107, and in run 7 the lighter model was slightly higher. The measurement is coarse: the models answer in roughly 0.05 steps, so with three personas a single persona moving one step shifts the run mean by about 0.017.

One further note on comparability: the double-counting probe passes a temperature of 0.7 to Sonnet, while Opus does not accept a temperature parameter and runs at its default. The two models are therefore not matched on sampling temperature. This is constant across every run.

Part C: the retracted-claim probe (covid_large), three runs per model

The probe plants one genuinely retracted claim (the withdrawn 2020 HIV-1-insertions preprint) as apparent lab-origin evidence, in two arms: shown live, and shown with an explicit retraction note. `retracted_still_moved` is how much belief still moved when the claim was labelled retracted; near zero means the reasoner correctly discounted withdrawn evidence.

Measure	Opus	Sonnet
mean_retracted_still_moved, run 1	0.00	-0.010
mean_retracted_still_moved, run 2	0.00	-0.010
mean_retracted_still_moved, run 3	0.00	-0.015
mean across runs	0.00	about -0.012
mean_retraction_ignored_effect	about 0.048	about 0.040

Reading: both models correctly discounted the retracted claim. Shown live, belief moved toward a lab origin by about 0.04 to 0.10; shown with the retraction, that movement essentially vanished. Two honest caveats: the retraction was explicitly labelled, so this shows models honoring a clear retraction rather than catching a quietly withdrawn claim; and Sonnet returned unparseable output on three cells across the three runs, so its means rest on fewer observations than Opus (which had none). Part A and Part B were unchanged in these runs: held-out transfer stayed 1.0 on every run, and the double-counting effect stayed in its established range.